

Chapter 13



Lean-Six Sigma Management & DMAIC Tools

Quality in Supply Chain Management

Opening Case: Welding Defects at a Saudi Refinery

Setting the Stage — Real-World Context

OPENING CASE

The Challenge

A Saudi downstream refinery was experiencing a 4.8% weld-failure rate on critical pipeline joints

- Nearly 10× the industry benchmark of 0.5%. Every rejected weld costs SAR 14,000 in rework and causes unplanned downtime.

How DMAIC Solved It:

Define → Charter: reduce weld rejects below 0.8%

Measure → Gauge R&R: found 22% measurement error

Analyze → Fishbone: root cause = gas-flow setting

Improve → Taguchi DOE: optimal pressure/temp/time

Control → SPC charts locked in the new settings

4.8%

initial weld-failure rate

0.6%

after DMAIC — SAR 2.1M saved/year



Opening Discussion

Before we begin — think about this:

"Think of a quality problem you have seen in a Saudi company or read about recently. What was the problem? How was it identified? Do you think a structured approach like DMAIC could have helped? Why or why not?"

 **Pair-Share — 2 minutes, then one spokesperson per pair reports back**

What Is Six Sigma?

Definition, Origins & the Core Equation

CONCEPT

Origins & Definition

σ (sigma) = Greek letter for standard deviation

Six Sigma = 6 standard deviations between process mean and nearest spec limit

Born at Motorola in 1982 when CEO demanded costs cut in half — twice.

Today: An organisation-wide improvement programme involving hierarchical training, analytical tools, and pay-for-knowledge.

The Core Equation

$$Y = f(X)$$

Y = output | X = inputs & process variables

Sigma	Long-run PPM
3 σ	66,807
4 σ	6,210
5 σ	233
6 σ	3.4

3-Sigma vs. 6-Sigma: A Visual Comparison

CONCEPT

3-Sigma Process

66,807 defects per million opportunities

- ~6.7% of products fail
- Mean shift causes rapid failure
- High rework & warranty costs
- Customer complaints frequent



6-Sigma Process

3.4 defects per million opportunities

- Near-zero defects
- Robust to process mean shifts
- Minimal rework & warranty
- Strong customer satisfaction

Example: Saudi telecom processes ~200M transactions/year. At 3 σ : 1.3M failures. At 6 σ : fewer than 680 failures.

Organising Lean-Six Sigma: The Belt Hierarchy

CONCEPT

Champion

Senior leader who sponsors projects, removes barriers, and allocates resources

Master Black Belt

Full-time trainer & mentor of Black Belts; oversees deployment strategy

Black Belt

Full-time project leader; completes cost-reduction projects (1–2 per year, \$200K+ savings each)

Green Belt

Part-time practitioner; applies basic tools and supports BB projects

Yellow Belt

Aware of Six Sigma concepts; participates in improvement teams

 *Black Belt training costs SAR 37,000–75,000 per person; expected project returns can reach SAR 750,000+ per project.*

Why Combine Lean with Six Sigma?

Waste Reduction + Variation Reduction

 CONCEPT

LEAN

Targets WASTE (Muda)

- Value Stream Mapping (VSM)
- Pull production / JIT
- 5S Workplace organisation
- Kaizen continuous improvement
- Kanban scheduling

+

SIX SIGMA

Targets VARIATION

- DMAIC structured process
- Gauge R&R (measurement)
- Taguchi DOE
- Statistical Process Control
- Capability analysis (Cpk)

Together: Lean-Six Sigma eliminates waste AND reduces variation — delivering speed, quality, and cost improvement simultaneously.

The DMAIC Roadmap

Five Phases — From Problem to Permanent Solution

 CONCEPT

D

DEFINE

Define project scope, goals & customer requirements

M

MEASURE

Measure current performance; collect baseline data

A

ANALYZE

Identify root causes of defects using data & tools

I

IMPROVE

Implement solutions; run controlled experiments

C

CONTROL

Sustain gains using SPC charts & control plans

DEFINE Phase: Building the Business Case

RUMBA Test for Project Selection

○ DEFINE

What Happens in DEFINE?

1. Identify a group of candidate projects
2. Write the business case (Objectives + Justification)
3. Perform risk & return analysis
4. Pareto analysis of Cost of Poor Quality (COPQ)
5. Write formal project definition (charter)

The RUMBA Test

R	Realistic:	Are goals attainable? Is the timeline feasible?
U	Understandable:	Is the case clearly written?
M	Measurable:	Are metrics defined?
B	Believable:	Is the projected saving credible?
A	Actionable:	Can it actually be implemented?

DEFINE: Project Risk & Return Matrix

Selecting the Right Project to Pursue

DEFINE

Risk Score = weighted sum of risk factors ÷ max possible score (%). Return Score = weighted sum of growth/urgency/impact ratings.

**HIGH-VALUE
TARGET**
Low Risk / High Return
Priority project

**STRATEGIC
BET**
High Risk / High Return
Rigorous analysis needed

**LOW-HANGING
FRUIT**
Low Risk / Low Return
Do quickly

AVOID
High Risk / Low Return
Drop it

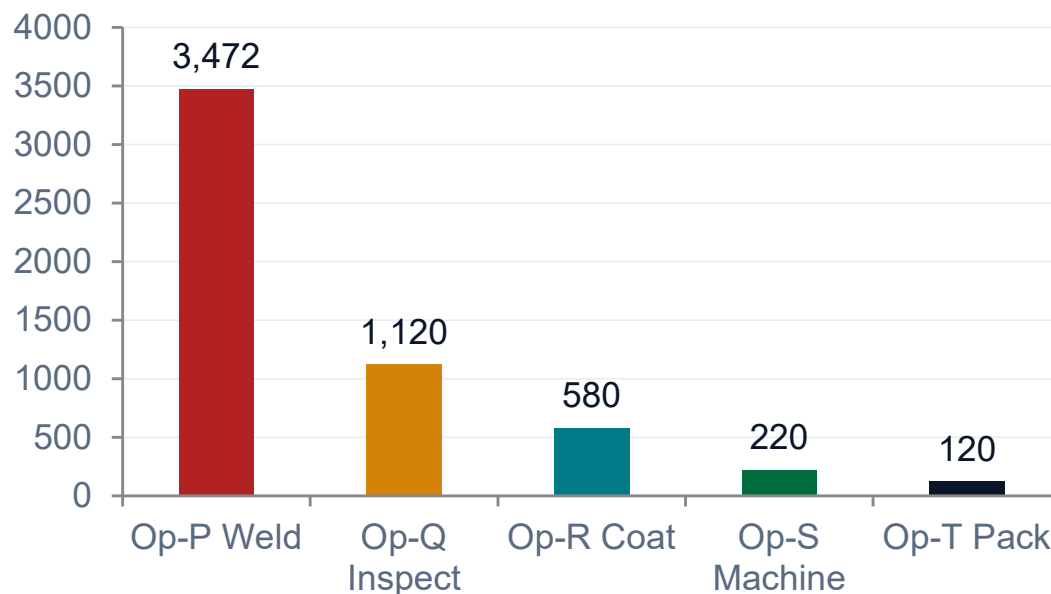
↑ HIGH
RETURN
↓ LOW

Our refinery case (Risk 39%, Return 40%) → Low-Hanging Fruit — execute quickly!

← LOW RISK — HIGH RISK →

DEFINE: Pareto Analysis & Project Charter

COPQ by Operation — Plant A (SAR '000)



Project Charter Elements

Problem Statement:	Weld failure rate of 4.8% vs 0.8% benchmark
Objective:	Reduce to $\leq 0.8\%$ within 6 months
Primary Metric:	Weld reject rate (%)
Secondary Metric:	Rework cost (SAR/month)
Scope:	Pipeline section B-12 welds only
Team:	BB + 2 GB + Welding Eng. + QC Mgr

MEASURE Phase: Selecting Outcomes & Verifying Data

You Can't Improve What You Can't Measure

MEASURE

Process Map

Flowchart with responsibility. Identifies non-value-added (NVA) steps. Tracks DPU & DPMO.

XY Matrix

Prioritises inputs (Xs) by their impact on outputs (Ys) using a weighted correlation grid.

FMEA

Failure Mode & Effects Analysis. Calculates Risk Priority Numbers (RPN) for potential failure modes.

Gauge R&R

Measures repeatability (same operator) and reproducibility (different operators) of measurement systems.

Capability Study

Computes Cp and Cpk to assess whether a process is capable of meeting customer specifications.

MEASURE: DPMO, DPU & Process Capability

MEASURE

Key Measurement Metrics

DPU

DPU = Defects ÷ Units

Defects per unit produced

DPMO

DPMO = (DPU ÷ Opportunities) × 1,000,000

Standardised to per-million scale

Cp

Cp = (USL – LSL) ÷ 6σ

Process spread vs. spec width

Cpk

Cpk = min[(USL–μ)/3σ, (μ–LSL)/3σ]

Accounts for mean offset

Worked Example

Scenario:

A pipe welding station produces 1,200 welds/day. Inspectors found 58 defects. Each weld has 3 inspection opportunities.

$$\text{DPU} = 58 \div 1200 = 0.048$$

$$\begin{aligned}\text{DPMO} &= (0.048 \div 3) \times 1,000,000 \\ &= 16,111 \text{ DPMO} \approx 3.6\sigma\end{aligned}$$

Target: reduce to <1,000 DPMO (≈4.6σ)

MEASURE: XY Matrix

Prioritising Process Inputs by Impact on Outputs

MEASURE

Rate each Input-Output relationship (1–10). Multiply by the output importance weight. Sum across rows → prioritised input list.

Input (X)	Weld Strength (wt=10)	Surface Finish (wt=7)	Porosity (wt=9)	Total Score	Priority
Gas flow rate	9	4	10	217	1st ↑
Electrode diam.	8	3	7	152	2nd
Travel speed	7	5	5	150	3rd
Pre-heat temp.	5	2	8	135	—
Arc voltage	6	3	6	123	—

Gas flow rate scored highest (217) → focus Analyze phase investigation here first.

ANALYZE Phase: Finding Root Causes

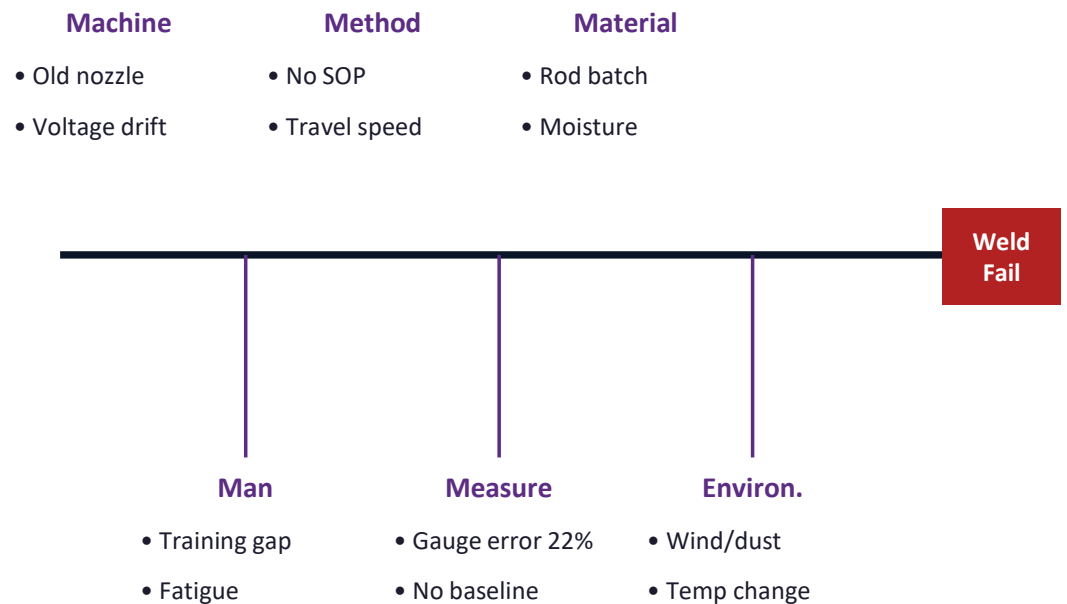
From Data to Diagnosis

ANALYZE

Key Analyze Tools

- Fishbone (Ishikawa) diagram
- 5-Why root cause analysis
- Scatter plots & regression
- Hypothesis testing (t-test, ANOVA)
- Capability analysis (Cp, Cpk)
- Failure Mode & Effects (FMEA)

Fishbone: Weld Failure Root Causes



ANALYZE: The 5-Why Technique

Drilling to the True Root Cause

● ANALYZE

Problem: Weld porosity failures — 4.8% reject rate at Saudi refinery pipeline section B-12.

Why 1

Q: Why are welds failing porosity test?

A: Because gas shield is inconsistent during welding

Why 2

Q: Why is the gas shield inconsistent?

A: Because gas flow rate fluctuates above 28 L/min

Why 3

Q: Why does gas flow rate fluctuate?

A: Because the regulator valve has a worn seal

Why 4

Q: Why is the regulator seal worn?

A: Because maintenance is done annually, not every 6 months

Why 5

Q: Why is maintenance interval 12 months?

A: Because no written SOP defined the correct interval for this gas type ← ROOT CAUSE



Fix: Create SOP mandating 6-month regulator maintenance. Assign responsibility to Lead Welder.

ANALYZE: Failure Mode & Effects Analysis (FMEA)

ANALYZE

RPN = Severity × Occurrence × Detection / RPN > 120: Immediate action required

Failure Mode	Effect	Sev	Occ	Det	RPN	Action
Gas flow fluctuation	Porosity, reject weld	8	7	4	224	Replace regulator seal
Worn electrode tip	Inconsistent arc	7	5	5	175	Tip-change checklist
Wrong travel speed	Incomplete fusion	8	4	5	160	Revise SOP + training
Pre-heat skipped	Cracking near toe	9	3	4	108	Visual check
Wrong filler alloy	Strength below spec	10	2	3	60	Storage labelling

IMPROVE Phase: Designing the Fix

Experiments, Solutions & Pilot Testing

IMPROVE

1. Generate Solutions

Brainstorm corrective actions for confirmed root causes. Use creativity tools (benchmarking).

2. Design of Experiments

Use Taguchi DOE or full factorial to find optimal settings for controllable process variables.

3. Evaluate & Select

Score solutions against criteria: impact, cost, feasibility, speed. Use a selection matrix.

4. Pilot the Solution

Run a controlled pilot on a small scale before full deployment. Verify gains are real.

5. Implement & Validate

Roll out at full scale. Confirm improvement against primary metrics defined in the charter.

IMPROVE: Taguchi Design of Experiments

Finding Optimal Settings Efficiently

IMPROVE

What is Taguchi DOE?

A structured approach to determine the best combination of input variables (factors & levels) to optimise a process output.

Key Concepts:

- Control factors: Variables management CAN control
- Noise factors: Uncontrollable variation
- Signal-to-Noise (S/N) ratio: Robustness measure
- Orthogonal Array: Efficient trial design

Objectives:

Smaller-the-better | Nominal-is-best | Larger-the-better

Taguchi vs. Full Factorial

Factors	Full Factorial	Taguchi L-Array
3 factors @ 2 levels	8 runs (2^3)	4 runs (L4)
4 factors @ 2 levels	16 runs (2^4)	8 runs (L8)
3 factors @ 3 levels	27 runs (3^3)	9 runs (L9)
4 factors @ 3 levels	81 runs (3^4)	9 runs (L9)

Taguchi DOE: Welding Optimisation (Worked Example)

Factors: A=Gas Flow (20/26 L/min), B=Arc Voltage (22/26V), C=Travel Speed (300/400 mm/min). Objective: Larger-the-better (weld strength score).

Trial	A Gas Flow	B Arc Voltage	C Travel Speed	Rep 1	Rep 2	Mean Score
1	A1 (20)	B1 (22V)	C1 (300)	72	68	70.0
2	A1 (20)	B2 (26V)	C2 (400)	78	82	80.0
3	A2 (26)	B1 (22V)	C2 (400)	75	77	76.0
4	A2 (26)	B2 (26V)	C1 (300)	88	86	87.0 ★

Factor Averages → A1 avg: 75.0 A2 avg: 81.5 → A2 better B1 avg: 73.0 B2 avg: 83.5 → B2 better C1 avg: 78.5 C2 avg: 78.0 → C1 better

✓ Optimal: A2 (26 L/min gas), B2 (26V arc), C1 (300 mm/min) — confirming experiment required.

IMPROVE: Taguchi Quality Loss Function (QLF)

Quantifying the Cost of Deviation from Target

IMPROVE

The QLF Concept

Traditional view: A product is either 'good' (within spec) or 'bad' (out of spec) — a step function.

Taguchi's view: ANY deviation from the target value causes loss — even within spec. Loss increases as a quadratic function of deviation.

Formula:

$$L(y) = K \times (y - T)^2$$

where $K = C_o / \Delta_o^2$ (C_o = repair cost, Δ_o = tolerance)

L = expected loss per unit

K = quality loss coefficient

V = mean squared deviation from target

Worked Example

Scenario:

Weld penetration tolerance: 6 ± 0.5 mm

Repair cost if rejected: SAR 14,000

Mean squared deviation: $(1/6)^2$ mm²

Solution:

$$K = 14,000 \div 0.5^2 = 56,000$$

$$L = 56,000 \times (1/6)^2 \\ = \text{SAR } 1,556 \text{ per weld}$$

CONTROL Phase: Sustaining the Gains

Preventing Backsliding After Improvement

● CONTROL

What Happens in CONTROL?

- 1. Statistical Process Control (SPC):** Implement X-bar & R charts on critical process parameters. Set control limits. React to special causes
- 2. Control Plan:** Document what to control, how often, who is responsible, and what to do when out of control.
- 3. Standardise & Document:** Update SOPs, training materials, and process documentation to reflect the new optimal settings.

Sample Control Plan

Parameter	Limit	Freq.	Owner	Response
Gas flow	26±1 L/m	Every 2h	Welder	Adjust regulator
Arc voltage	26±0.5V	Per job	Welder	Check machine
Travel speed	300±20	Daily	QC Tech	Recalibrate
Tip condition	Visual	Per shift	Foreman	Replace tip

Lean Tool: Value Stream Mapping (VSM)

Seeing Waste Before You Eliminate It

LEAN TOOL

What Is VSM?

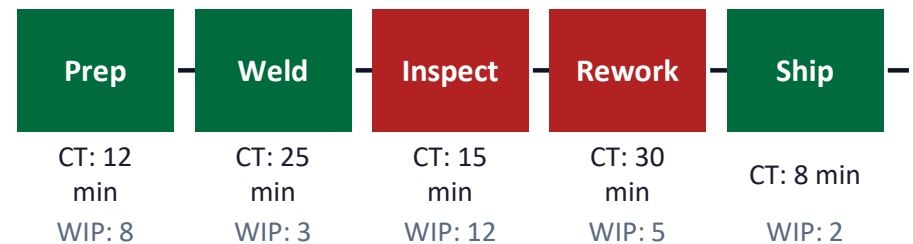
A VSM shows the entire material and information flow for a product family — from supplier to customer.

Key Metrics in VSM:

- Lead Time (total elapsed time)
- Value-Added Time (actual processing)
- Process cycle efficiency = $VA\ Time \div Lead\ Time$
- Work-in-Process (WIP) at each stage

Lean target: Eliminate NVA steps (transport, waiting, overproduction, defects, inventory, motion, over-processing)

Simplified VSM — Pipe Welding Cell



Total LT: 90 min | VA Time: 45 min | PCE: 50% → Target: >75%
by eliminating inspect/rework loop

Legend:  VA (Value-Adding)  NVA (Non-Value-Adding) CT = Cycle Time WIP = Work-in-Process

Why Lean-Six Sigma Fails

Critical Success Factors from a Contingency Perspective

✗ Lack of Champion leadership

→ Projects stall without resources or management support

✓ Champion must attend reviews and actively remove barriers

✗ Wrong culture — no tolerance for failure

→ Teams hide problems instead of surfacing them

✓ Psychological safety + celebrate learning from pilots

✗ Lack of data / poor measurement system

→ Decisions made on noise, not signal

✓ Gauge R&R before any analysis; don't proceed without it

✗ Misunderstood roles & responsibilities

→ BB/GB confusion leads to duplicated or missed work

✓ RACI matrix defined at project kickoff

✗ Resistance to change

→ Operators revert to old methods after control phase

✓ Involve operators in IMPROVE; make them co-owners

✗ Wrong project selection

→ Resources spent on low-impact projects

✓ Use risk-return matrix to prioritise portfolio

DMAIC Tools Quick Reference Guide

The Analyst's Toolkit — One Phase at a Time

Phase	Primary Tools	Key Output / Deliverable	Saudi Industry Example
DEFINE	Project charter, RUMBA, Pareto, risk-return matrix	Signed project charter with clear metrics	Aramco identifies top-10 COPQ sources across 5 refineries
MEASURE	Process map, XY matrix, FMEA, Gauge R&R, DPMO	Prioritised input list + verified measurement system	SABIC validates pellet weighing scale has <10% R&R
ANALYZE	Fishbone, 5-Why, scatter plots, hypothesis testing	Confirmed root cause(s) with statistical evidence	Almarai proves sealing temp deviation causes 68% of rejects
IMPROVE	Taguchi DOE, orthogonal arrays, QLF, brainstorming	Optimal process settings from designed experiment	Saudi Steel finds A2+B2+C1 settings → strength +22%
CONTROL	SPC (X-bar, R, p-charts), control plan, SOPs	Sustained gains documented in control plan	NEOM project locks weld settings with SPC dashboard

Common Mistakes — And How to Avoid Them

Lessons from Real Six Sigma Deployments

⚠ MISTAKES & FIXES

Gauge R&R

- ✗ *Skipping R&R and proceeding with data collection*
- ✓ Always verify measurement system first — if R&R>30%, all data is suspect

FMEA

- ✗ *Computing RPN but never updating after improvements*
- ✓ Re-run FMEA post-improvement to verify RPN reduction

Control Phase

- ✗ *Declaring victory after pilot — no control plan implemented*
- ✓ DMAIC is not done until SPC charts are live and operators are trained

XY Matrix

- ✗ *Assigning correlation scores without data — pure opinion*
- ✓ Use historical data or a pilot study to validate the ratings

Taguchi DOE

- ✗ *Applying a learning curve to noise factors instead of control factors*
- ✓ Classify each factor as control or noise before designing the array

5-Why

- ✗ *Stopping at symptom level (Why 2-3) and missing root cause*
- ✓ Keep asking until you reach a systems-level answer; verify with data

Decision Guide: Which Tool for Which Situation?

 You need to select the right Six Sigma project	Risk & Return Matrix + Pareto of COPQ
 You need to understand which inputs drive outputs	XY Matrix (Measure phase)
 Your measurement system might be unreliable	Gauge Repeatability & Reproducibility (Gauge R&R)
 You need to identify why a process is failing	Fishbone + 5-Why + FMEA (Analyze phase)
 You need to find the optimal process settings	Taguchi Design of Experiments (Improve phase)
 You want to quantify the cost of deviation from target	Quality Loss Function (QLF)
 You want to sustain improvements long-term	SPC Charts + Control Plan (Control phase)
 You are designing a new product or process	Design for Six Sigma (DFSS: DMADV or IDOV)

Exit Ticket — Before You Leave

Consolidate Your Learning

EXIT ACTION

1. Name the 5 DMAIC phases

...and give ONE tool used in each phase.

2. What is DPMO?

...and what sigma level corresponds to 6,210 DPMO?

3. Explain in one sentence

...why Taguchi DOE is more efficient than full factorial.

4. When would you use DFSS instead of DMAIC?

...give an example from a Saudi industry context.

5. Pose ONE question

...you still have after today's lecture.

 Write answers on a notecard and submit — or post to the class LMS discussion board.

Key Takeaways

- 1 Six Sigma uses a data-driven DMAIC roadmap to reduce defects to 3.4 per million opportunities.
- 2 Lean and Six Sigma are complementary — Lean attacks waste, Six Sigma attacks variation.
- 3 The Measure phase is critical: bad data from a poor measurement system invalidates all analysis.
- 4 The XY matrix and FMEA prioritise where to invest analytical effort during Analyze.
- 5 Taguchi DOE finds optimal settings with far fewer experiments than full factorial — a strategic advantage.
- 6 Control phase is where gains are won or lost: no control plan = no sustainable improvement.
- 7 DFSS (DMADV/IDOV) is used for NEW products; DMAIC is used to fix EXISTING processes.
- 8 Saudi Vision 2030 manufacturing ambitions require Lean-Six Sigma as a core competence.

Next Steps, Assignments & Final Challenge

Applying DMAIC Beyond the Classroom

ACTION ITEMS

Individual Assignment

Select a real Saudi company facing a quality problem. Apply the DMAIC framework: write the business case, identify 5 Xs using an XY matrix, draw a fishbone, and propose a Taguchi L4 experiment. (2–3 pages, due next lecture)

Team Project Milestone

Complete the Measure phase for your term project process: submit process map, XY matrix with ranked inputs, and a Gauge R&R study plan. Present Week 11.

Recommended Reading

Foster, Chapter 13 (full). Supplement: Montgomery 'Introduction to Statistical Quality Control', Ch. 11 (Taguchi Methods) & George, Rowlands, Price, Maxey 'Lean Six Sigma Pocket Toolkit'.

Final Reflection Challenge

You are hired as a Black Belt at a NEOM construction logistics company. Defect rate in prefabricated wall panels is 4.1%. Design the entire DMAIC roadmap — which 3 tools would you deploy in each phase, in what sequence, and why?